Roll No. ....

TCE–703

B. TECH. (CE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

TRANSPORTATION ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Compare rail transport with road transport, listing advantages and disadvantages of both. (CO1)

OR

(b) Discuss the uni-gauge policy of Indian Railways. Describe the benefits of the uni-gauge system. (CO1)

2. (a) Define railway track or a permanent way. Describe the requirements of an ideal permanent way. (CO1)

OR

(b) List out the various gauges prevailing in India with their gauge widths. Discuss the various factors that govern the selection of a suitable gauge. (CO1)

P. T. O.

3. (a) Calculate the maximum permissible load that a BG locomotive with three pairs of driving wheels bearing an axle load of 22 each can pull on a straight level track at a speed of 80 km/h. Also calculate the reduction in speed if the train has to run on a rising gradient of 1 in 200. What would be the further reduction in speed if the train has to negotiate a 4° curve on the rising gradient ? Assume the coefficient of friction to be 0.2. (CO1)

OR

(b) Describe the requirements of an ideal rail joint. Discuss the problems that arise due to rail joints. (CO2)

4. (a) Discuss briefly the causes and effects of creep in the railway track. (CO2)

OR

(b) Explain the functions of rails. How is the weight of a rail section usually determined ? (CO2)

5. (a) Describe the requirements of sleepers used in a railway track. Explain the functions of sleepers. (CO2)

OR

(b) Define ballast. Why is it used in the railway track ? Briefly describe the various types of ballasts used. (CO2)

60